9 **(a)** State the principle of conservation of momentum.

.....

.....

..... [2]

(b) Two frictionless trolleys A and B with attached identical springs approach each other along a horizontal line, as illustrated in Figure 9.1.

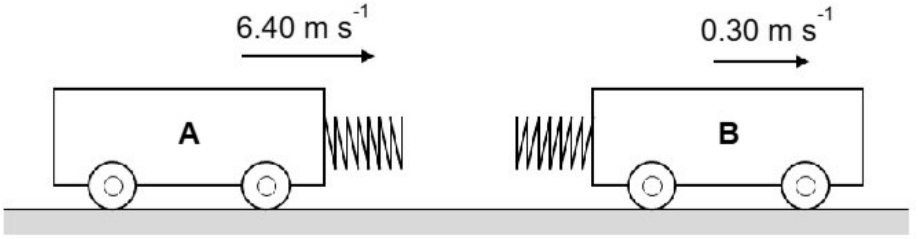


Figure 9.1

Trolley A has mass 1.2 kg and a velocity of 6.40 m s^{-1}. Trolley B has mass 5.2 kg and a velocity of 0.30 m s^{-1}.

The two trolleys collide elastically. After collision, trolley A reverses its direction of motion.

(i) Show that the speed of trolley B after the collision is 2.6 m s^{-1}.

[3]

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(ii) Figure 9.2 shows the velocity-time graph of trolley B.

Sketch also on Figure 9.2, the velocity-time graph for trolley A and label it as **A**.

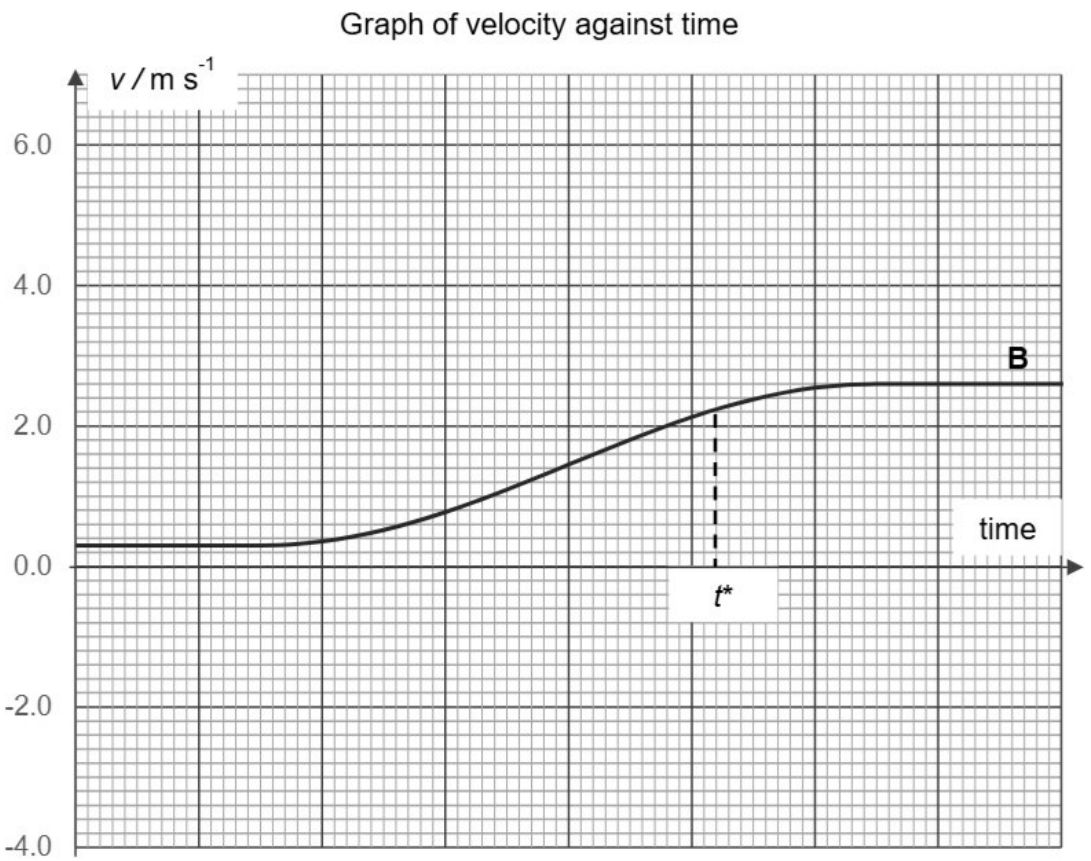


Figure 9.2

[3]

(iii) Explain how the contact force on trolley B by trolley A at time *t** can be determined from Figure 9.2.

.....

.....

..... [2]

(iv) Describe the change in the total kinetic energy of the system of trolleys A and B throughout the collision.

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..... [2]

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(v) Calculate the maximum elastic potential energy stored in *each* spring.

maximum elastic potential energy = ... J [3]

(c) After the collision of the two trolleys, trolley B moves to the edge of the platform and falls through a vertical height of 10.0 m before being brought to rest in a duration of 0.100 s, without rebounding.

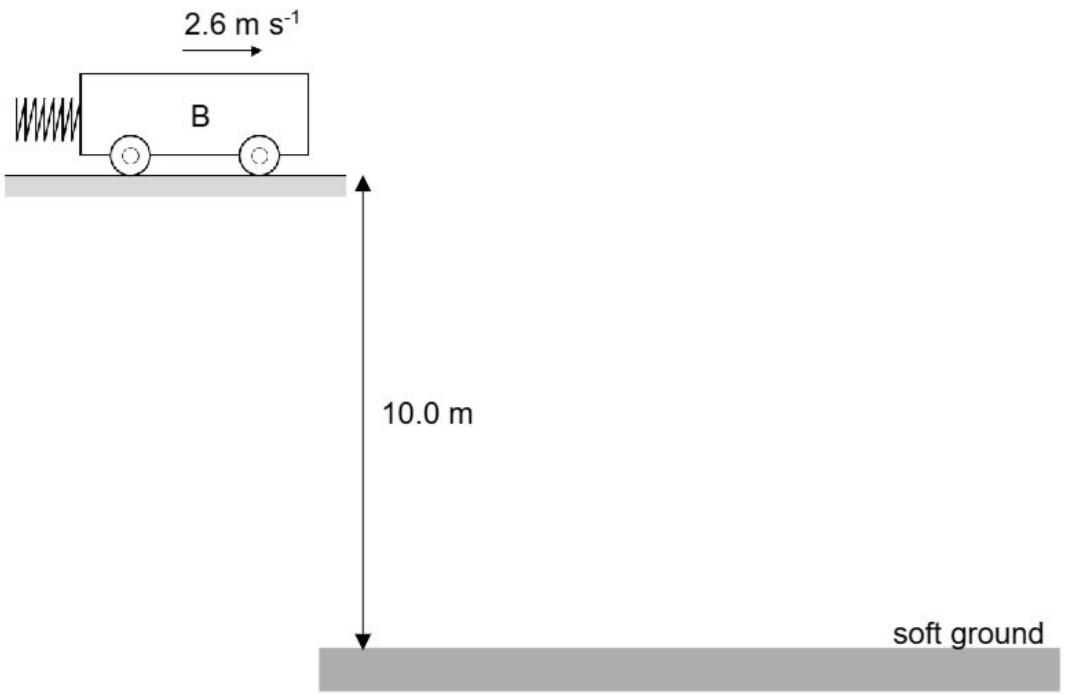


Figure 9.3 (not to scale)

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(i) Determine the average vertical contact force acting on the trolley by the soft ground.

force = ... N [4]

(ii) Explain what happens to the impulse on the trolley if it were to land on hard ground and rebounded instead.

.....

..... [1]

[Total: 20]